

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js workflow** for MCA filings with session handling, CAPTCHA handoff, retries, and state recovery, reducing manual filing time by **85%**.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized backend services with Docker, wired GitHub Actions deployments to Google Cloud Run, and improved release velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling for **95 brain subparts**, handling preprocessing, orientation correction, model inference, generated masks, and large-volume data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged complex pipeline issues across conformation, orientation, label mapping, PyTorch/ONNX-style inference, and binary mask generation in a mature open-source codebase.

Open Source

[Contributions](#) 

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

Projects

Swish Support System | *Python, Go, FastAPI, Gin, PostgreSQL, Redis, Langfuse*

[GitHub](#) 

- Built AI workflow infrastructure for support conversations with context memory, policy-grounded action selection, multimodal evidence checks, and traceable LLM assessment outputs.
- Separated semantic understanding from deterministic business rules so ambiguous or high-risk cases could be logged, reviewed, and handled through safe fallbacks instead of prompt-only decisions.

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, yfinance, JSONL*

- Built a multi-agent research workflow with role-specific analysts, debate, risk review, portfolio approval, checkpointed state, and decision memory for repeated market windows.
- Added replayable traces, validation harnesses, and scorecards so agent behavior, memory use, and decision quality could be inspected instead of treated as a black box.

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Built a self-hosted **AI security gateway** for LLM, HTTP-tool, and MCP traffic with route profiles, request scoring, runtime policy checks, and allow/block decisions before downstream execution.
- Added RBAC runtime settings, PII redaction, audit logs, attack-lab validation, OpenTelemetry traces, Prometheus metrics, and health checks for policy and failure visibility.

Technical Skills

Languages: Python, Go, TypeScript, C++

Frameworks: FastAPI, Gin, Echo, Node.js, Agno, PyTorch

Tools: PostgreSQL, Redis, Docker, Kubernetes, AWS, GCP, OpenTelemetry, Prometheus, Langfuse, GitHub Actions